

Two Sets II

Time limit: 1.00 s

Memory limit: 512 MB

Your task is to count the number of ways numbers $1, 2, \dots, n$ can be divided into two sets of equal sum.

For example, if $n = 7$, there are four solutions:

- $\{1, 3, 4, 6\}$ and $\{2, 5, 7\}$
- $\{1, 2, 5, 6\}$ and $\{3, 4, 7\}$
- $\{1, 2, 4, 7\}$ and $\{3, 5, 6\}$
- $\{1, 6, 7\}$ and $\{2, 3, 4, 5\}$

Input

The only input line contains an integer n .

Output

Print the answer modulo $10^9 + 7$.

Constraints

- $1 \leq n \leq 500$

Example

Input	Output
7	4